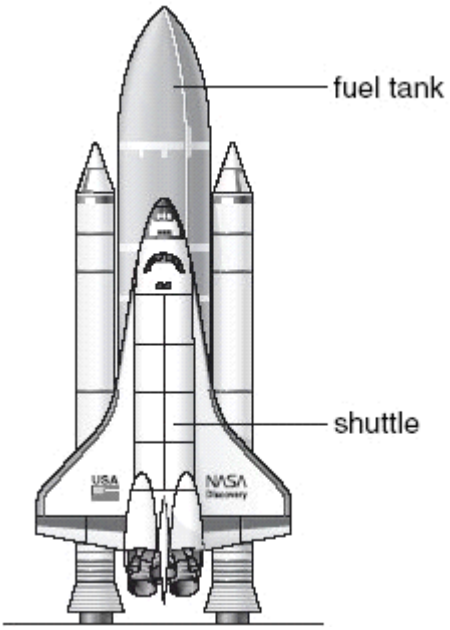
Q1. The shuttle is a spacecraft which is used to take satellites into space.
The drawing below shows the shuttle just about to take off.



(a) The shuttle has a separate fuel tank containing liquid hydrogen and liquid oxygen.

Explain why hydrogen and oxygen are transported as liquids rather than as gases.

.....

.....

1 mark

(b) Oxygen is needed to burn the fuel in the shuttle's engines.
Vehicles on Earth do **not** need a tank containing oxygen.

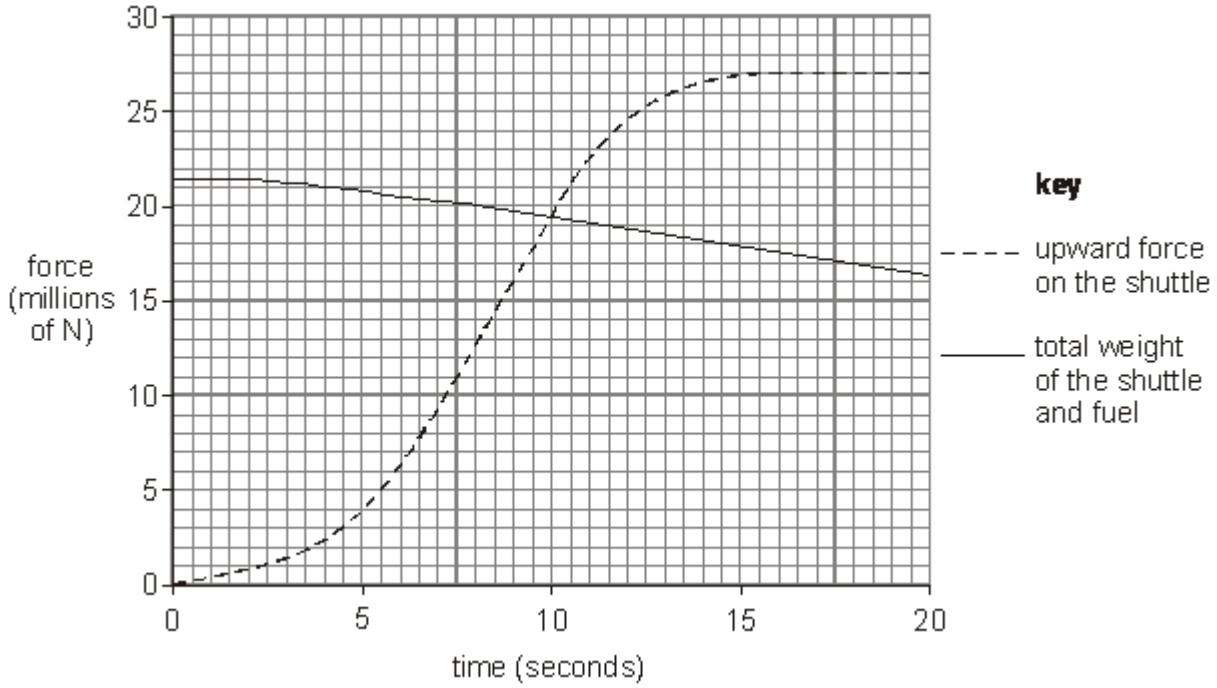
Why does the shuttle need to have a tank containing oxygen?

.....

.....

1 mark

(c) The graph below shows how the upward force and the weight of the shuttle, including fuel, change during the first 20 seconds, after the fuel is ignited.



Why does the total weight of the shuttle **decrease** during the first 20 seconds?

.....

.....

1 mark

(d) (i) Look at the graph. At 20 seconds, what is the value of:

the upward force on the shuttle?

..... millions of N

the total weight of the shuttle and fuel?

..... millions of N

1 mark

(ii) At 20 seconds, what is the **resultant** force on the shuttle?

..... millions of N

1 mark

(iii) Use the graph to explain why the shuttle **cannot** take off before 10 seconds.

.....

.....

1 mark

maximum 6 marks

Q1.

(a) any **one** from

- liquids **or** they take up less space
  accept 'particles are closer together in a liquid'
- liquids **or** they are denser
  accept 'particles are further apart in a gas'
  *do **not** accept liquids are heavier'*
  'liquids are easier to transport' is insufficient

1 (L7)

(b) there is no oxygen in space
accept 'there is no air in space'
accept 'space is a vacuum'

1 (L7)

(c) any **one** from

- the weight of the fuel decreases
  'gravity gets weaker' is insufficient
- there is less fuel on board
- fuel is being used up
  *accept 'hydrogen and oxygen **or** oxygen is being used up'*
- water is released
  accept 'waste gas is given off'

1 (L7)

(d) (i)
- 27
  ***both** answers are required for the mark*
- 16.5
  accept a number from 16.3 to 16.7

1 (L6)

(ii) 10.5
*accept the difference between the numbers given in **d i***
consequential marking applies

1 (L7)

(iii) any **one** from

- up to 10 seconds the weight is greater than the (upward) force
- after 10 seconds there is a resultant **or** net (upward) force
- after 10 seconds the (upward) force is greater than the weight
  accept the converse

answers must be in terms of a comparison

'it does not have a great enough upward force' is insufficient

'at 10 seconds the forces are equal' is insufficient

1 (L7)

[6]